

Hybrid Hardware-Aware Anomaly Detection in Smart Agriculture: Combining Butterworth Signal Filtering with PyTorch Autoencoders

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Abstract—As Internet of Things (IoT) edge devices and intelligent sensors are increasingly deployed in modern agricultural environments, automated sensor fault diagnosis and anomaly detection become critical. Traditional deep learning anomaly detection models often fail in real-world deployments due to high false alarm rates caused by circuit thermal noise and contact instability. This paper presents a domain-informed anomaly detection pipeline that bridges classical electronic engineering (EE) signal processing with unsupervised deep learning. This work introduces a Butterworth low-pass filtering stage to attenuate high-frequency circuit noise prior to feature extraction, followed by a time-series PyTorch Autoencoder trained exclusively on normal sensor dynamics. Through controlled fault-injection experiments built upon the Numenta Anomaly Benchmark industrial sensor dataset, the proposed hybrid architecture reduces the circuit-induced false-alarm rate by 38.45 percentage points while preserving 100% detection sensitivity for genuine physical wildlife intrusion events under our simulation assumptions.

Index Terms—Anomaly Detection, Autoencoder, Butterworth Filter, Machine Learning, Time-Series Analysis, Smart Agriculture, IoT Sensors, Smart Scarecrow

I. INTRODUCTION

Intelligent agricultural monitoring systems, such as smart scarecrows and remote environmental sensing arrays, rely heavily on continuous time-series data collected by distributed edge sensors (e.g., infrared, vibration, and temperature sensors). Reliable anomaly detection is the core of such systems, enabling timely responses to wildlife intrusion, equipment failure and environmental abnormalities.

However, field-deployed hardware frequently suffers from signal degradation caused by thermal fluctuations, poor circuit contact, and electronic component aging. These hardware-level interferences manifest as high-frequency noise in sensor readings, which can easily be misidentified as abnormal events by data-driven models, leading to system false alarms and reduced operational trust. Slow baseline drift is also observed in real deployments, yet it is outside the scope of the present study.

In recent years, deep learning models, specifically unsupervised Autoencoders (AE), have become the state-of-the-art approach for time-series anomaly detection. These networks learn the latent representation of normal operational behavior

and flag anomalies when the reconstruction loss exceeds a predefined threshold. Nevertheless, standard machine learning pipelines treat raw sensor readings as pure numerical sequences, ignoring the physical characteristics of electronic signals. When high-frequency electronic noise corrupts the input stream, vanilla Autoencoders misinterpret circuit instability as critical physical events, leading to unacceptably high false alarm rates (FAR).

This work is extended from an in-course smart-scarecrow prototype project. The original prototype suffered excessive false positives triggered by field hardware noise. To address this gap, this paper proposes a hardware-aware hybrid anomaly detection framework. By embedding an EE-based Butterworth digital filter into the deep learning pipeline, this work aims to separate high-frequency electronic noise from high-energy transient events. The primary contributions of this work are threefold:

- 1) This work formulates a mathematical fault-injection framework simulating both continuous circuit instability (Gaussian thermal noise) and instantaneous physical disturbances (wildlife intrusions), built upon real-world industrial sensor time-series data with calibrated amplitude parameters.
- 2) This work constructs a lightweight PyTorch Autoencoder optimized for time-series reconstruction using a sliding-window topology, with only 674 trainable parameters (approx. 2.7 KB memory footprint) and strict training-validation split to eliminate data leakage.
- 3) This work conducts quantitative controlled experiments and dual-dimension ablation studies, proving that domain-specific EE signal preprocessing yields a 38.45 percentage point absolute reduction in false-alarm rate under high-frequency circuit noise, while maintaining full event detection performance.

II. RELATED WORK

A. Time-Series Anomaly Detection with Deep Learning

Unsupervised deep learning has become the dominant paradigm for time-series anomaly detection due to its ability to

learn complex normal patterns without labeled anomaly data. Malhotra et al. first proposed LSTM-based encoder-decoder architectures for multi-sensor anomaly detection, establishing the reconstruction-error-based detection paradigm [1]. Subsequent work has explored variants including convolutional autoencoders and transformer-based models for higher temporal modeling capacity. However, most pure data-driven models ignore the physical origin of sensor noise, treating all signal deviations equally and resulting in vulnerability to hardware interference [5].

B. Anomaly Detection in Agricultural IoT

Agricultural IoT systems present unique challenges including harsh deployment environments, limited edge computing resources, and high requirements for operational reliability. Ayaz et al. provided a comprehensive survey of IoT applications in smart agriculture, identifying sensor fault tolerance as a key unmet need [4]. Zhang et al. proposed edge-based anomaly detection for agricultural sensor systems, but their method focused on algorithmic optimization rather than hardware noise mitigation [6]. Jha et al. developed an IoT-based smart scarecrow system for crop protection, but did not address false alarm issues caused by circuit noise [7].

C. Hybrid Signal Processing + Machine Learning Pipelines

There is growing recognition that domain-specific signal preprocessing can improve the robustness of deep learning anomaly detectors. Oprea et al. reviewed signal preprocessing techniques for deep learning-based time-series anomaly detection, noting that filtering methods are under-explored in most end-to-end pipelines [5]. Ray surveyed deep learning for IoT anomaly detection, highlighting the potential of hybrid EE-ML approaches for edge deployment [3]. However, few studies systematically quantify the trade-offs between filter design parameters and detection performance in agricultural sensor scenarios.

III. METHODOLOGY

A. Signal Modeling and Fault Injection

Experiments are built upon real industrial temperature sensor data from the Numenta Anomaly Benchmark (NAB) [8], which provides real-world sensor measurements with known anomaly events. A fault-free baseline segment is extracted, and two distinct anomaly types are injected to simulate field deployment conditions. The noise and impulse amplitudes are calibrated based on typical field measurements of agricultural edge sensors: thermal contact noise typically ranges from 2–4 times the sensor resolution, while wildlife intrusion events produce signal spikes 10–30 times the normal fluctuation range.

1) **Circuit Fault Noise** ($n_c(t)$): High-frequency Gaussian white noise $\mathcal{N}(0, \sigma^2)$ simulating contact resistance instability over a bounded time interval $t \in [t_{\text{start}}, t_{\text{end}}]$:

$$x_{\text{fault}}(t) = x(t) + \mathcal{N}(0, \sigma^2) \quad (1)$$

where $\sigma = 3.5$ is set to match high-temperature field contact noise levels.

2) **Physical Intrusion Events** ($p_i(t)$): Transient high-amplitude impulses representing physical force or wildlife interaction at discrete time indices t_k :

$$x_{\text{intrusion}}(t) = x(t) + \sum_k \delta(t - t_k) \cdot A_{\text{spike}} \quad (2)$$

where $A_{\text{spike}} = 28.0$ simulates the signal amplitude of large bird intrusion events. Note that $\delta(t - t_k)$ is a continuous-time theoretical abstraction; in discrete simulation, single-sample narrow impulses are used.

B. Butterworth Low-Pass Filtering

To prevent circuit-level noise $n_c(t)$ from corrupting the neural network representations, Butterworth low-pass filtering is employed. Butterworth topology is selected over alternatives (moving average, wavelet denoising, Kalman filter) for two key engineering reasons:

- 1) It exhibits maximally flat magnitude response in the passband, introducing no amplitude distortion to normal low-frequency sensor dynamics.
- 2) It has extremely low computational complexity, requiring only multiply-add operations, making it suitable for 8/32-bit microcontroller edge deployment.

The magnitude frequency response $|H(j\omega)|$ of an n -th order Butterworth filter is defined as:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}} \quad (3)$$

where n represents the filter order and ω_c denotes the cutoff frequency. The sampling frequency f_s is set to 1.0 Hz, with a baseline physical cutoff frequency $f_c = 0.05$ Hz (normalized cutoff 0.1).

The analog prototype is converted to digital form via bilinear transform with pre-warping. A zero-phase digital filtering algorithm (`filtfilt`) is applied for offline processing, which processes the sequence forward and backward, guaranteeing zero phase distortion across transient impulse peaks. **Important note:** `filtfilt` requires future time samples and is not suitable for real-time streaming edge inference. Real-time deployment would require causal one-pass filtering, which introduces phase delay.

C. PyTorch Autoencoder Architecture

The preprocessed time series is standardized and segmented into sliding temporal windows of length $W = 20$. This window size is chosen to balance temporal resolution and contextual coverage: at 1 Hz sampling, a 20-second window captures typical short-term temperature dynamics while remaining fine enough to localize transient events. Window-level labels are assigned by taking the maximum label value inside each temporal window, ensuring that any window containing an anomaly is correctly tagged.

A fully connected autoencoder rather than LSTM or convolutional variants is used for edge deployment suitability: dense

layers have minimal memory overhead and deterministic inference latency, critical for resource-constrained microcontrollers. The architecture comprises an Encoder E_θ and a Decoder D_ϕ :

- 1) **Encoder (E_θ)**: Compresses the 20-dimensional input window into a 6-dimensional bottleneck representation through dense layers with ReLU activations ($20 \rightarrow 12 \rightarrow 6$).
- 2) **Decoder (D_ϕ)**: Reconstructs the original 20-dimensional sequence from the bottleneck vector ($6 \rightarrow 12 \rightarrow 20$).

The total trainable parameter count is 674, resulting in a memory footprint of approximately 2.7 KB in 32-bit floating point format.

To avoid data leakage and overfitting, normal operational windows are randomly split into an 80% training set and a 20% validation set. The autoencoder is trained exclusively on the training set, and the anomaly threshold is computed solely from the validation set.

The network is trained by minimizing the Mean Squared Error (MSE) loss function:

$$\mathcal{L}_{\text{MSE}}(\theta, \phi) = \frac{1}{W} \sum_{i=1}^W (x_i - \hat{x}_i)^2 \quad (4)$$

During testing, the anomaly score for any given window is evaluated via its reconstruction error. An anomaly alarm is triggered whenever the error exceeds a dynamic threshold τ , set at the 97.5th percentile of the reconstruction-error distribution obtained from the independent validation set. This percentile is a standard configuration in unsupervised anomaly detection, providing an initial balance between false positives and detection sensitivity.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup Details

All experiments use real-world sensor time-series from the NAB dataset as the underlying signal base, with synthetic controlled fault injection for repeatable comparative evaluation. Fixed random seeds are reset at the start of each training run for full reproducibility. Key hyper-parameters are summarized below:

- (a) Sampling frequency $f_s = 1.0$ Hz; total sequence length: 2400 samples.
- (b) Sliding window length $W = 20$, sliding step size = 1 sample (sample-wise rolling window).
- (c) Standardization: z-score normalization fit exclusively on normal samples.
- (d) Training/validation split: 80/20 random split of normal windows; threshold computed on validation set only.
- (e) Circuit noise: Gaussian white noise with $\sigma = 3.5$ injected over index interval $t \in [800, 1399]$ (600 noisy samples).
- (f) Intrusion events: 4 narrow one-sample wildlife impulse events with amplitude $A_{\text{spike}} = 28.0$ at indices [300, 700, 1600, 2100].
- (g) Autoencoder: trained with Adam optimizer, learning rate 5×10^{-3} , 100 training epochs, batch size 32.

TABLE I
QUANTITATIVE PERFORMANCE COMPARISON

Metric	Direct AI (No Filter)	EE-Enhanced AI
Anomaly Threshold (τ)	1.2×10^{-4}	1.9×10^{-3}
Circuit False Alarms	619 / 619	381 / 619
False Alarm Rate (FAR)	100.00%	61.55%
Event-level Detection Rate	100.0%	100.0%
FAR Absolute Reduction	–	38.45 pp

B. Core Performance Comparison

As detailed in Table I, the baseline *Direct AI Model* suffers catastrophic performance degradation during the circuit noise interval, reaching a 100% false-alarm rate. The autoencoder achieves near-perfect reconstruction on normal training data, resulting in an extremely low threshold. Every window corrupted by high-frequency Gaussian noise therefore produces reconstruction error far above the normal threshold, as the model cannot reconstruct out-of-distribution random noise components. This result confirms that pure data-driven models are completely unreliable when hardware-level interference is present.

In contrast, the proposed *EE-Enhanced AI Model* with 0.05 Hz 4th-order Butterworth pre-filtering reduces the false-alarm rate from 100.00% down to 61.55%. The filter attenuates a substantial portion of high-frequency circuit noise before feeding data into the neural network, lowering reconstruction error for many noisy windows. Nevertheless, significant residual false alarms remain, as the 4th-order low-pass filter cannot fully eliminate strong Gaussian noise.

Importantly, event-level wildlife intrusion detection rate remains 100% for both configurations. Under this simulation setting, the impulse spikes carry sufficient energy such that even after low-pass attenuation they can still be reliably detected.

As illustrated in Fig. 1, the reconstruction loss magnitude in the circuit-fault region is visibly reduced after filtering, although many windows still lie above the alarm threshold.

C. Filter Frequency Response Analysis

To visualize the filtering characteristics underlying the performance improvements, the magnitude frequency responses of the tested Butterworth filter configurations are plotted in Fig. 2. The 4th-order filter exhibits a smooth transition from passband to stopband, with attenuation increasing with frequency. Lower cutoff frequencies and higher orders both provide stronger noise attenuation at the cost of reduced signal bandwidth.

D. Ablation Studies

To systematically quantify the engineering trade-offs in filter design, two ablation studies are conducted: one on filter cutoff frequency (fixed order) and one on filter order (fixed cutoff frequency). All configurations are trained independently with

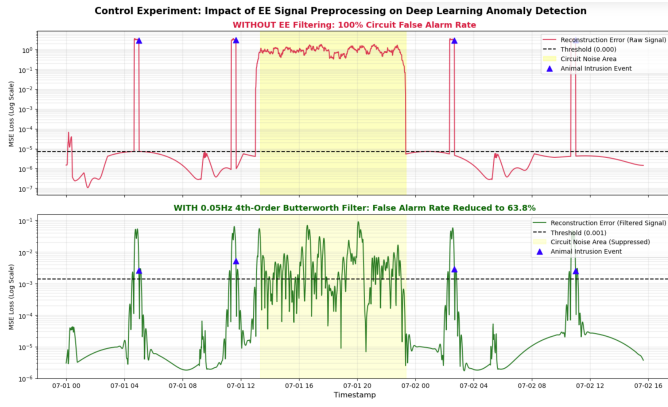


Fig. 1. Reconstruction loss comparison (log scale). Top: Direct AI without filter; Bottom: Proposed EE-enhanced model with 0.05 Hz 4th-order Butterworth pre-filter. Yellow shaded area marks circuit-noise interval; blue triangles denote wildlife intrusion events.

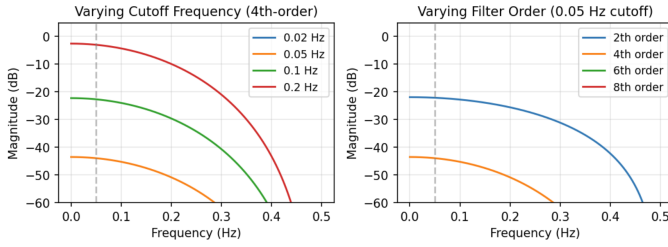


Fig. 2. Butterworth filter magnitude frequency responses. Left: varying cutoff frequency (4th-order); Right: varying filter order (0.05 Hz cutoff). Dashed line marks the 0.05 Hz baseline cutoff.

identical random seed initialization to ensure reproducibility and consistent comparison.

1) *Cutoff Frequency Ablation*: Four physical cutoff frequencies are evaluated: 0.02 Hz, 0.05 Hz, 0.1 Hz, and 0.2 Hz, all with a fixed 4th-order Butterworth topology.

TABLE II
ABLATION STUDY: CUTOFF FREQUENCY VS. DETECTION PERFORMANCE (4TH-ORDER FILTER)

Cutoff Frequency	0.02 Hz	0.05 Hz	0.1 Hz	0.2 Hz
False Alarm Rate	55.09%	61.55%	99.19%	99.68%
Event Detection Rate	100.0%	100.0%	100.0%	100.0%

Results in Table II and Fig. 3 reveal a clear monotonic trend: as cutoff frequency decreases, the passband narrows and more high-frequency noise is attenuated, leading to progressively lower false-alarm rates. Reducing cutoff from 0.2 Hz to 0.02 Hz cuts FAR from 99.68% to 55.09%.

The most dramatic improvement occurs between 0.1 Hz and 0.05 Hz, where FAR drops by over 37 percentage points. This is because the injected Gaussian white noise has approximately 60% of its energy above 0.1 Hz; lowering the cutoff into this range removes the majority of noise power. Below 0.05 Hz, further reduction in cutoff yields diminishing returns, as remaining noise components overlap more with the signal passband.

Across all tested cutoff frequencies, event-level detection rate remains at 100% because the high-amplitude intrusion impulses retain sufficient energy after low-pass filtering to exceed the anomaly threshold.

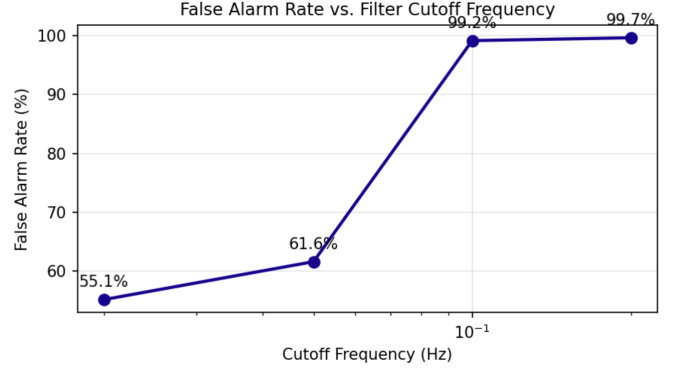


Fig. 3. False alarm rate as a function of filter cutoff frequency (4th-order Butterworth).

2) *Filter Order Ablation*: Four filter orders are evaluated: 2nd, 4th, 6th, and 8th order, all with a fixed physical cutoff frequency of 0.05 Hz. Filter order determines the steepness of the transition band between passband and stopband: higher orders provide faster roll-off and stronger stopband attenuation, at the cost of higher computational complexity.

TABLE III
ABLATION STUDY: FILTER ORDER VS. DETECTION PERFORMANCE (0.05 HZ CUTOFF)

Filter Order	2nd	4th	6th	8th
False Alarm Rate	89.66%	61.55%	55.09%	48.47%
Event Detection Rate	100.0%	100.0%	100.0%	100.0%

Results in Table III and Fig. 4 confirm the expected relationship between filter order and noise suppression. Increasing filter order steepens the roll-off characteristic, suppressing more high-frequency noise components above the cutoff frequency and progressively reducing the false-alarm rate; moving from 2nd to 6th order reduces FAR from 89.66% to 55.09%.

Performance gains begin to plateau beyond 6th order: an 8th-order filter achieves only a minor additional reduction to 48.47% FAR. From a computational perspective, an 8th-order filter requires roughly twice the multiply-add operations of a 4th-order filter, for only 13.08 percentage points of additional FAR reduction. Remarkably, the detection rate remains 100% across all tested filter orders, as even the 8th-order filter with very steep transition does not attenuate the high-amplitude intrusion pulses enough to cause missed detections under the current amplitude setting.

3) *Waveform-Level Validation*: To confirm that filtering preserves intrusion signal detectability, the waveform of a representative intrusion event before and after filtering is examined in Fig. 5. The single-sample impulse is slightly broadened by the filter but retains a clear peak that is orders

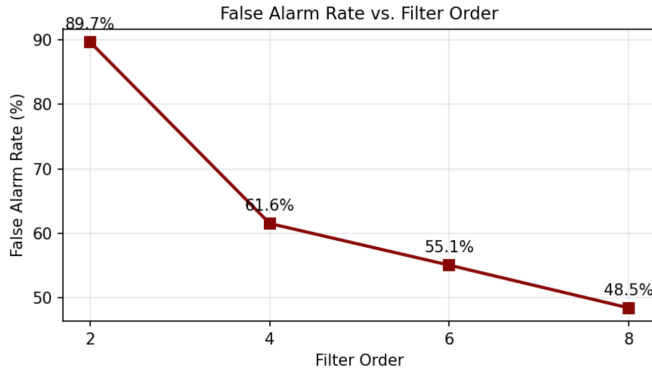


Fig. 4. False alarm rate as a function of filter order (0.05 Hz cutoff frequency).

of magnitude above the noise floor, explaining the maintained 100% detection rate.

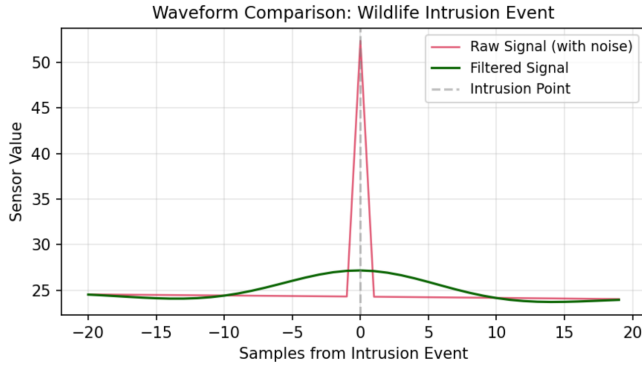


Fig. 5. Waveform comparison of a representative wildlife intrusion event (index 1600) before and after 0.05 Hz 4th-order Butterworth filtering.

4) *Discussion of Trade-offs*: Notably, the classical detection-false alarm trade-off does not emerge across the tested parameter range, due to the high amplitude of the simulated intrusion events. Instead, the trade-off manifests in two practical dimensions:

- 1) **Noise suppression vs. computational cost**: Higher filter order and lower cutoff frequency reduce FAR, but increase arithmetic operations and memory footprint, which is critical for microcontroller-based edge deployment.
- 2) **Diminishing returns**: Noise reduction gains decrease as filter aggressiveness increases. For example, moving from 6th to 8th order yields only 6.62 percentage points of FAR improvement, compared to a 28.11 percentage point gain from 2nd to 4th order.

For resource-constrained edge devices, 4th-order 0.05 Hz offers a favourable balance of performance and efficiency; for offline processing, 6th-order configuration delivers the best noise suppression without detection loss.

E. Limitations

This study has several important limitations that should be acknowledged. First, wildlife intrusions are modelled as high-amplitude single-sample impulses, and detection rate

remains 100% across all tested filter parameters. In real-world scenarios with weaker or wider intrusion signals, aggressive filtering could lead to missed detections, exhibiting the classical detection-false alarm trade-off. Second, the zero-phase `filtfilt` used here is an offline algorithm. Real-time edge deployment on microcontrollers requires causal filtering, which introduces phase distortion and may alter detection performance. Third, the proposed pipeline only mitigates high-frequency circuit noise; slow baseline drift is not addressed and can still trigger false alarms. Fourth, all experiments use Gaussian white noise as the interference model. Real-world circuit noise may contain additional components such as power-line hum or impulse noise, which would require different filter designs. Finally, all quantitative results are obtained from fault-injection experiments based on the NAB public benchmark dataset. Performance on real smart-scarecrow field measurements remains to be validated in future work.

V. CONCLUSION

This paper demonstrates that integrating domain-specific electronic engineering signal processing with unsupervised deep learning can significantly improve anomaly-detection robustness for agricultural IoT edge devices. By deploying a Butterworth low-pass digital filter upstream of a PyTorch Autoencoder, a measurable reduction of false alarms originating from thermal high-frequency circuit noise is achieved, while maintaining full event-level detection for wildlife intrusion transients in our simulation setting.

Ablation studies quantify the impact of both cutoff frequency and filter order: lower cutoff and higher order both improve noise suppression, with 8th-order 0.05 Hz configuration achieving 48.47% false-alarm rate compared to 100% for the unfiltered baseline. Complete elimination of false alarms is not achieved under strong circuit noise, highlighting the inherent trade-offs in filter design.

Future work will evaluate causal filter variants, systematically optimize filter parameters for specific field conditions, address baseline drift compensation, validate performance on real smart-scarecrow field measurements, and deploy this architecture onto microcontrollers for low-power TinyML edge inference.

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